

**A REPORT ON
(Blogging Platform)
By**

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CERTIFICATE

This is to certify that the project entitled Blogging Platform has been successfully completed by **Neeharika Veluvolu - AP24110011072, Mohanasritha Eerla- AP24110011024, Neelima Bojanapu - AP24110011111, Madhurya Meduri - AP24110011107, Geethika Marrivada - AP24110010971** as a required academic component of the course CSE 211 – Full Stack Development during Semester 4 of Academic Year 2025-26 in the Department of Computer Science and Engineering at SRM University, AP. This work was executed under the guidance of the undersigned and is deemed to have successfully met all course requirements as of 27-04-2026

(Signature of the Course Instructor)

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1.INTRODUCTION:

BlogIt is a modern and user-friendly full-stack blogging platform designed to simplify the creation, management, and interaction of digital blog content. It enables users to write, edit, and publish rich-text blog posts using an intuitive interface. The platform supports features such as categories, tags, likes, bookmarks, and comments, making it highly interactive and engaging.

Users can explore posts across multiple categories like Technology, Education, Science, and more. BlogIt also provides a powerful search functionality using full-text indexing, allowing users to quickly find relevant content. The platform ensures secure authentication using JWT and maintains data integrity through structured database management.

With additional features like a community discussion page and a unique typewriter writing mode, BlogIt offers both productivity and creativity. Its flexibility and scalable architecture make it suitable for students, writers, and developers looking for a complete blogging solution.

2.SCENARIO BASED CASE STUDY:

Meet Rahul, a college student and aspiring writer who wants to share his ideas and technical knowledge with others. However, he struggles to find a simple platform where he can write, publish, and manage his blogs efficiently.

Rahul discovers BlogIt, a blogging platform designed to streamline content creation and interaction.

User Registration and Authentication:

Rahul registers on BlogIt and logs in securely using JWT-based authentication, ensuring his data is protected.

Blog Creation and Management:

He writes blog posts using a rich text editor, adds images, tags, and categories, and publishes them for others to read.

Content Exploration:

Rahul browses posts from different categories and uses the search feature to find blogs related to his interests.

User Interaction:

He likes posts, bookmarks useful articles, and comments on blogs to engage with other users.

Profile Management:

Rahul updates his profile with a bio and avatar, showcasing his identity as a writer.

Community Engagement:

He participates in discussions on the community page, sharing ideas and interacting with other users.

Rahul's Experience:

With BlogIt, Rahul can easily manage his blogging activities, improve his writing skills, and connect with a community. The platform helps him grow as a content creator and express his ideas effectively.

3.SYSTEM REQUIREMENTS:

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Node.js (v16 or above): Runtime environment for backend and frontend
- npm (v8 or above): Package manager for dependencies
- React.js: Frontend UI development
- Express.js: Backend API framework
- MongoDB: NoSQL database for storing users, posts, and comments
- Browser: Google Chrome / Firefox (latest version)
- Axios: HTTP client for API communication
- Visual Studio Code: Code editor

Hardware Requirements

- Processor: Intel Core i5 (8th Gen or above)
- RAM: Minimum 8 GB
- Storage: At least 1 GB free space
- Display: 1366x768 or higher

4.PROJECT ARCHITECTURE:

The BlogIt application follows a client-server architecture:

Frontend: Handles UI and user interactions

Backend: Processes requests and business logic

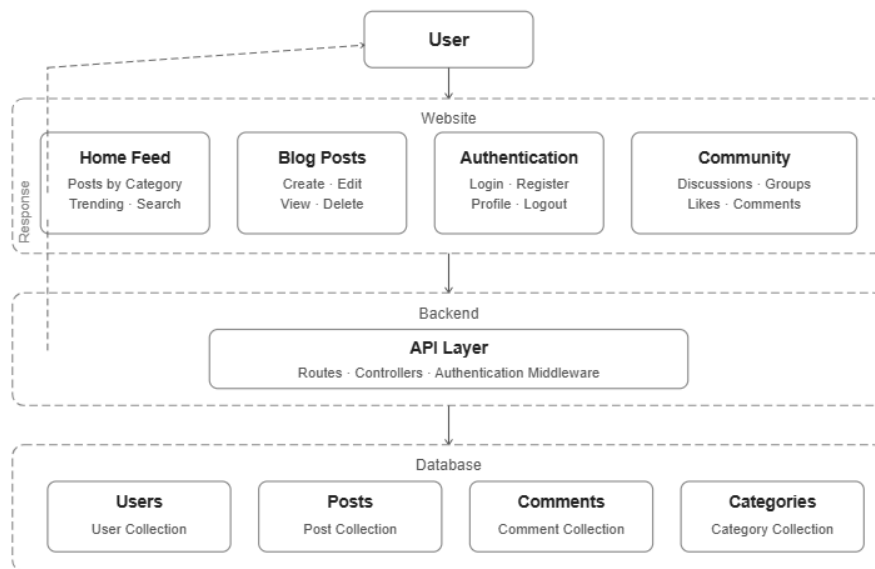
Database: Stores user data, posts, comments, and categories

5. TECHNICAL ARCHITECTURE:

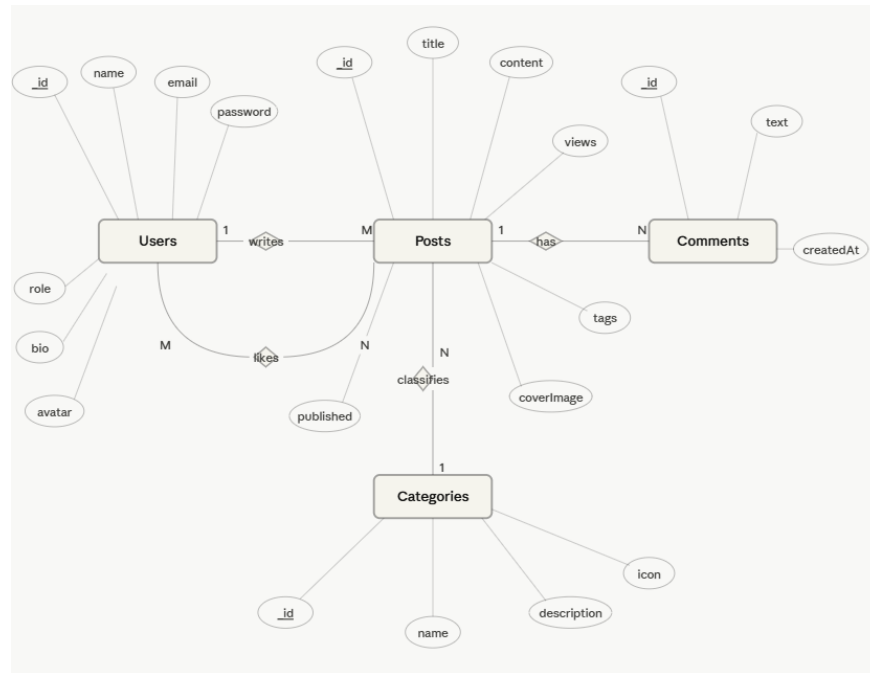
The system begins with the user interacting with the BlogIt frontend interface.

The frontend sends requests to the backend server through API calls using Axios. The backend processes these requests, performs necessary operations, and communicates with the MongoDB database to store or retrieve data.

The database stores information such as users, blog posts, comments, likes, and bookmarks. After processing, the backend sends responses back to the frontend, which updates the UI accordingly.



6.ER DIAGRAM:



Entities:

- **User:** Stores user details like name, email, password, bio, avatar
- **Post:** Contains blog content, title, category, tags, likes, views
- **Category:** Represents blog categories
- **Comment:** Stores comments on posts

Relationships:

- User creates Posts
- User likes bookmarks Posts
- Post belongs to Category
- Post has Comments
- User writes Comments

Primary Keys:

Each entity has a unique ID (User ID, Post ID, Category ID, Comment ID)

Foreign Keys:

- Post references User (author)
- Comment references User and Post

7.KEY FEATURES:

- User Registration and Authentication
- Blog Creation with Rich Text Editor
- Categories and Tag Management
- Likes and Bookmarks System
- Comments on Posts
- Full-Text Search Functionality
- Trending Posts (based on views and likes)
- User Profile Management
- Community Discussion Page
- Typewriter Writing Mode

8.ROLES AND RESPONSIBILITIES:

User:

- Register and manage account
- Create, edit, and delete blog posts
- Like, bookmark, and comment on posts
- Search and explore content
- Manage profile details

Admin:

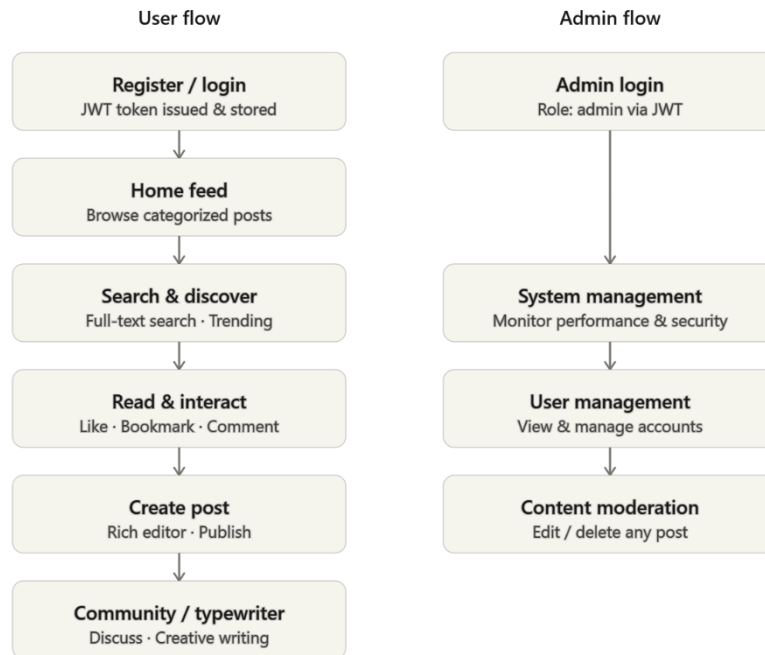
- Manage users and posts
- Monitor platform activity
- Ensure data security and system performance

9.USER FLOW:**User Flow**

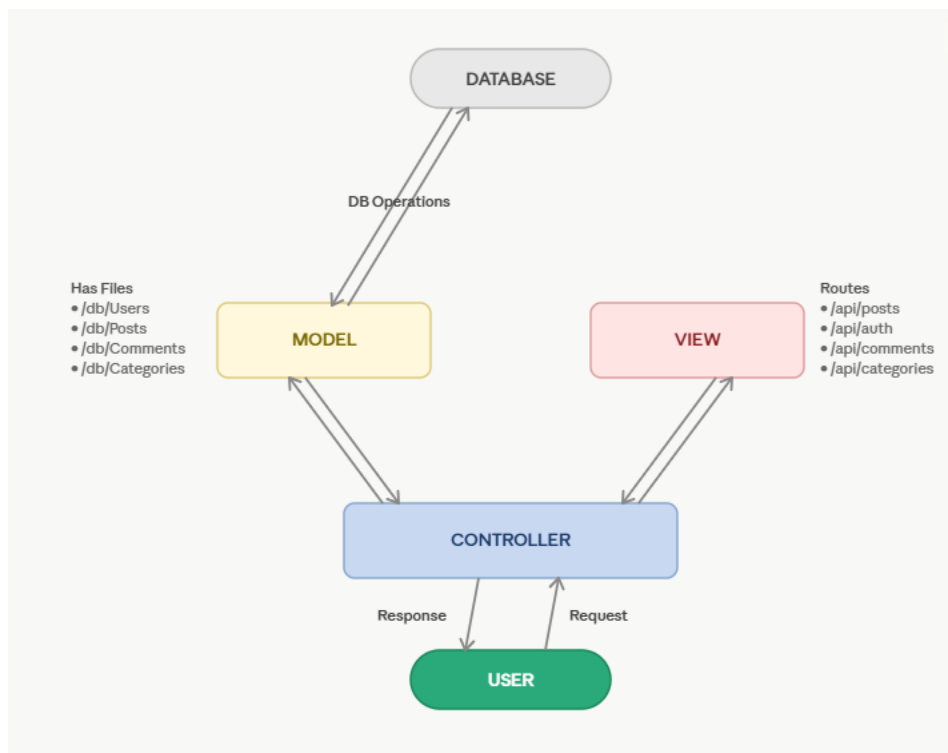
- Register/Login
- Browse or Search Blogs
- Create/Edit Blog Posts
- Like, Bookmark, Comment
- View Profile

Admin Flow

- Manage Users
- Manage Posts
- Monitor System



10.MVC PATTERN:



The BlogIt application follows the MVC architecture:

Model Layer

Handles database schemas using Mongoose (Users, Posts, Comments, Categories)

Controller Layer

Processes requests, applies logic, interacts with models

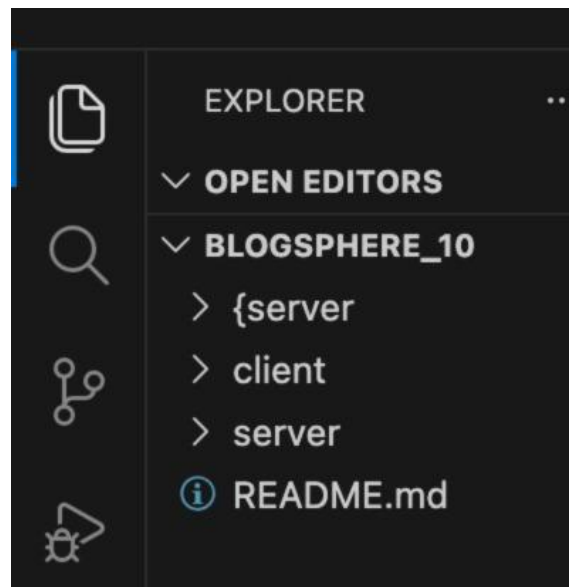
View Layer (Routes)

Defines API endpoints and handles HTTP requests

Advantages:

- Separation of Concerns
- Scalability
- Reusability
- Easy Testing
- Better Collaboration

11.PROJECT SETUP AND CONFIGURATION:



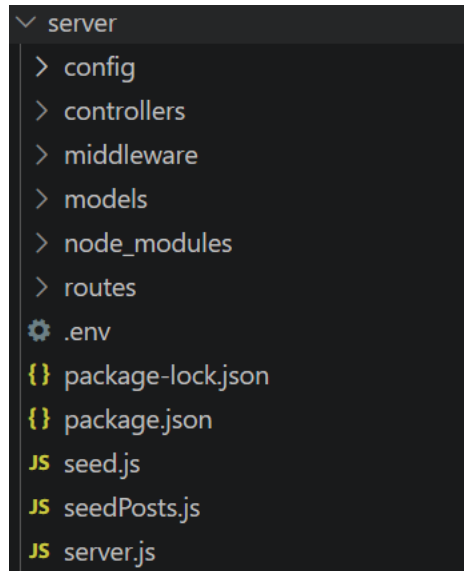
Client Setup

- Create React app using Vite
- Install dependencies using npm
- Run using npm run dev

Server Setup

- Initialize Node project using npm init
- Create folders: models, controllers, routes
- Run backend using npm run dev

12.BACKEND DEVELOPMENT:



Controllers

- authController - Handles login/register
- postController - Handles blog operations
- commentController - Handles comments
- userController - Handles user profile

Middleware

- authMiddleware - JWT authentication

Models

- Users
- Posts
- Comments
- Categories

Routes

- Auth Routes
- Post Routes

- Comment Routes
- User Routes

13.DATABASE DEVELOPMENT:

- MongoDB used as database
- Mongoose for schema creation

Schemas:

- User Schema
- Post Schema
- Comment Schema
- Category Schema

14.FRONT-END DEVELOPMENT:

Pages

- Login, Register
- Home
- Blog Detail
- Categories
- Create/Edit Post
- Profile
- Community
- Typewriter
- Search

Components

- Navbar
- Footer

- PostCard
- AuthContext

15.CONCLUSION:

BlogIt is blogging platform that successfully integrates modern web technologies to provide a seamless content creation and interaction experience. The system allows users to create, publish, and manage blog posts efficiently while also enabling engagement through likes, bookmarks, and comments.

The implementation of secure authentication using JWT, structured database design with MongoDB, and a responsive frontend using React ensures both performance and usability. Additional features like full-text search, trending posts, and categorized content enhance the overall user experience.

Overall, BlogIt demonstrates a scalable and modular architecture following the MVC pattern, making it easy to maintain and extend. It serves as an effective solution for individuals seeking a reliable and interactive blogging platform.

16.FUTURE SCOPE:

Although BlogIt provides a strong foundation, several enhancements can be implemented to further improve its functionality:

- Integration of real-time features such as live comments and notifications using WebSockets
- Development of a mobile application version for wider accessibility
- Enhanced security measures such as input validation and rate limiting
- Making the community discussion feature fully database-driven
- Adding recommendation systems based on user interests using AI/ML techniques
- Integration of social media sharing features

These improvements can make BlogIt more robust, scalable, and industry-ready.

17.REFERENCES:

- React.js Official Documentation – <https://react.dev/>
- Node.js Official Documentation – <https://nodejs.org/>
- Express.js Documentation – <https://expressjs.com/>
- MongoDB Documentation – <https://www.mongodb.com/docs/>
- JSON Web Tokens (JWT) – <https://jwt.io/>